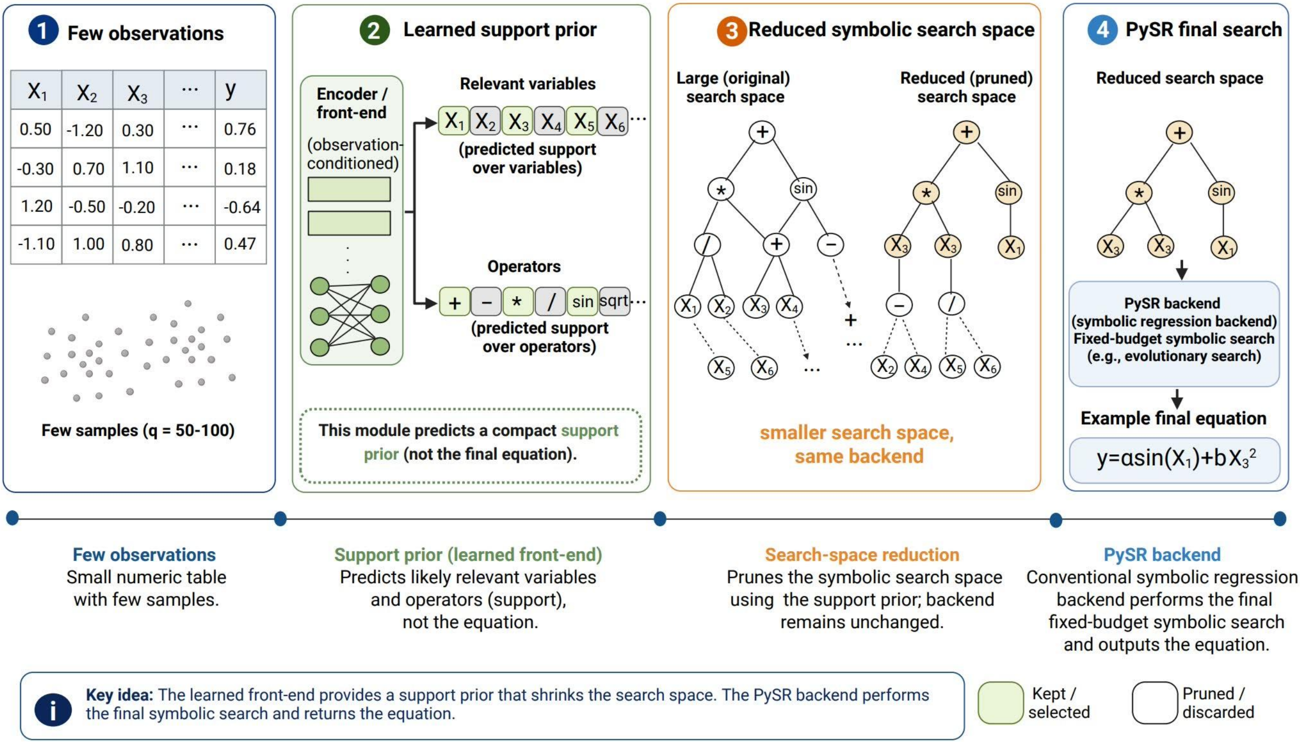

1 Few observations
X1 X2 X3 ··· y
0.50 -1.20 0.30 ··· 0.76
-0.30 0.70 1.10 ··· 0.18
1.20 -0.50 -0.20 ··· -0.64
-1.10 1.00 0.80 ··· 0.47
Few samples (q = 50-100)
2 Learned support prior
Encoder / front-end
(observation-conditioned)
Relevant variables
X1 X2 X3 X4 X5 X6 ···
(predicted support over variables)
Operators
+ − * / sin sqrt ···
(predicted support over operators)
This module predicts a compact support prior (not the final equation).
3 Reduced symbolic search space
Large (original) search space
Reduced (pruned) search space
+ * sin / + − X1 X2 X3 X4 X5 X6 ···
+ * sin X3 X3 X1 − / X2 X4 X5 X6
smaller search space, same backend
4 PySR final search
Reduced search space
+ * sin X3 X3 X1
PySR backend
(symbolic regression backend)
Fixed-budget symbolic search
(e.g., evolutionary search)
Example final equation
y=αsin(X1)+bX3²
Few observations
Small numeric table with few samples.
Support prior (learned front-end)
Predicts likely relevant variables and operators (support), not the equation.
Search-space reduction
Prunes the symbolic search space using the support prior; backend remains unchanged.
PySR backend
Conventional symbolic regression backend performs the final fixed-budget symbolic search and outputs the equation.
Key idea: The learned front-end provides a support prior that shrinks the search space. The PySR backend performs the final symbolic search and returns the equation.
Kept / selected
Pruned / discarded